Mannix Graham

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EDUCATION

Wilfrid Laurier University, Waterloo, ON

Bachelor of Science, Computer Science (Honours Specialization) - Expected Dec 2027

Relevant Coursework: Data Structures 1 and 2, Software Engineering, Object-Oriented Programming, Computer Networks, Discrete Mathematics, Artificial Intelligence, Data Algorithms, Ethics & Professional Practice

Certifications: AWS Certified Cloud Practitioner (CLF-C02) — Currently Preparing

TECHNICAL SKILLS

Cloud & DevOps: AWS (EC2, VPC, IAM, CloudWatch, CloudTrail), Terraform (IaC), CI/CD pipelines, GitHub Actions, Azure DevOps, Docker, Kubernetes (k3s), cloud cost optimization (FinOps)

Languages & Scripting: Python, Bash, PowerShell, Git/GitHub, REST APIs

Testing & Automation: Playwright, Selenium, pytest, deployment validation, log analysis, troubleshooting

Networking & Monitoring: CloudWatch, Power BI reporting, Cisco IOS (VLANs, inter-VLAN routing), site-to-site VPN, Proxmox VE, Tailscale, Ubuntu Linux

RELEVANT EXPERIENCE

Quality & Product Support Intern | Alcohol and Gaming Commission of Ontario, Toronto, ON

Python · Playwright · Selenium · Azure DevOps · Power BI · Beautiful Soup | May 2026 - Aug 2026

- Automated release-readiness smoke testing by leading R&D of the team's first Playwright (Python) suite across 14 test cases, **cutting full regression execution from 3.5 hours (manual) to 7 minutes, a 97% reduction**.
- Enabled the QA team to adopt automation by authoring a training guide and leading a live demo, **resulting in 4 of 6 teammates independently building and maintaining test cases post-internship**.
- Validated defect fixes with developers via log analysis in Azure DevOps, **resolving 1 defect and 5 support tickets weekly within 1–2 sprint cycles** and running 10+ test cases per sprint with no coverage gaps across 5–6 sprints.
- Validated operational readiness by executing disaster-recovery test scenarios against a 24-hour RTO across 3 critical systems, **documenting outcomes to support AGCO's business-continuity testing process**.
- Reduced manual monitoring effort with a Python scraper that tracks AGCO-related news and social mentions and surfaces volume trends, **later adopted by the AGCO Communications intern**.

PROJECTS

AI Cloud Cost Optimizer — FinOps Project

AWS Billing & Usage Data · AI Agent · Terraform · Human-in-the-Loop Approval

- Building an AI-driven tool that analyzes AWS billing and usage data to flag idle and oversized resources and recommend savings, with every infrastructure change gated by human approval, **turning cloud waste into measurable budget savings**.

Enterprise Hybrid AWS Environment — Cloud Portfolio Project

Terraform · AWS (VPC, EC2, IAM, CloudWatch) · GitHub Actions (OIDC) · Proxmox · Tailscale

- Provisioned a multi-AZ AWS VPC, EC2, IAM, and CloudWatch monitoring with modular Terraform (bootstrap, modules, environments), **enabling repeatable, console-free infrastructure builds**.
- Secured CI/CD deployments through GitHub Actions pipelines using short-lived OIDC credentials, **eliminating long-lived AWS access keys and reducing the environment's credential attack surface**.
- Extending the environment with a site-to-site VPN to a home-lab server to demonstrate hybrid cloud/on-prem networking and observability.

Self-Managed Network & Virtualization Homelab

Cisco IOS · Proxmox VE · Tailscale · Pi-hole · Ubuntu Linux · Docker · Kubernetes

- Designed and deployed a VLAN-segmented home network with inter-VLAN routing on Cisco IOS, isolating device traffic by security zone to reduce lateral-movement risk.
- Deployed a self-managed k3s Kubernetes cluster running Dockerized workloads on Proxmox-hosted VMs, with Tailscale for secure remote access, **demonstrating cloud-native operations without public exposure**.

ADDITIONAL WORK EXPERIENCE

Operations Associate | City of Waterloo | Sept 2025 – Present

- Maintained continuity of service for 200+ patrons per shift by troubleshooting incidents across facility systems and coordinating event setups with teams of 6–8 staff.

Facility Attendant | City of Brampton | Jan 2023 – Present

- Prevented unplanned equipment outages by monitoring ice, brine, and compressor systems each shift and escalating trend deviations for early incident response.